Rational Numbers: Converting Recurring Decimal to fraction

25/7/25
Day 5

* bar (‾) represents Recurring decimal

Ex:- $0.\overline{3} = 0.333\ldots$

$0.\overline{50} = 0.505050\ldots$

$1.\overline{03} = 1.0303\ldots$

$2.01\overline{4} = 2.0144\ldots$

1) $0.\overline{a} = \dfrac{a}{9}$

2) $0.\overline{ab} = \dfrac{ab}{99}$

3) $0.\overline{abc} = \dfrac{abc}{999}$

4) $0.a\overline{b} = \dfrac{ab - a}{90}$

+0 next to 9

5) $0.a\overline{bc} = \dfrac{abc - a}{990}$

6) $ab.cd$

*) Bar → Denominator → 9 (priority)

*) No Bar → 0 → (priority)

*) $ab.\overline{cd} = \dfrac{abcd - ab}{99}$

only Bar

Ex:- $23.\overline{43} = \dfrac{2343 - 23}{99} = \dfrac{2320}{99}$

or, normal

$x = 23.4343 \ldots$ ①

repeating 2 times $\times 100$ both side, for 1 time 10, for 3 time 1000.

$\therefore 100x = 2343.4343\ldots$ ②

$99x = 2343.4343 - 23.4343$

$x = \dfrac{2343 - 23}{99} = \dfrac{2320}{99}$

① Find the value of $0.\overline{3} + 0.\overline{6} + 0.\overline{7} + 0.\overline{8}$:

A) $2\frac{3}{10}$

B) $2\frac{2}{3}$

C) 2.35

D) $5\frac{3}{10}$

Sol^n The above q^n having single bars

$$= \frac{3}{9} + \frac{6}{9} + \frac{7}{9} + \frac{8}{9}$$

$$= \frac{3+6+7+8}{9}$$

$$= \frac{\cancel{24}\,8}{\cancel{9}\,3} = \frac{8}{3} \text{ (Improper)}$$

3 → small

$\frac{3}{8}$ (proper)

8 → big.

$$= \frac{8}{3}$$

$$= 2\frac{2}{3}$$ → mixed convert

D - 3) 8 (2 - Q
6
2 → R

$\boxed{Q\frac{R}{D} = 2\frac{2}{3}}$

a) $2\frac{3}{10}$

~~b)~~ $2\frac{2}{3}$

c) 2.35

d) $5\frac{3}{10}$

Ans b)

② The difference of $5.\overline{76}$ & $2.\overline{3}$ is :

a) $2.\overline{54}$

b) $3.\overline{73}$

c) $3.\overline{46}$

~~d)~~ $3.\overline{43}$

$$\begin{array}{r} 5.7676 \\ -\ 2.3333 \\ \hline 3.4343 \end{array}$$

Ans d)

③ The simplification of $3.\overline{36} - 2.\overline{05} + 1.\overline{33}$ is

a) $2.\overline{06}$

b) $2.\overline{61}$

~~c)~~ $2.\overline{64}$

d) $2.\overline{64}$

$$\begin{array}{r} 3.3636 \ldots \\ -\ 2.0505 \ldots \\ \hline 1.3131 \ldots \\ +\ 1.3333 \ldots \\ \hline 2.6464 \end{array}$$

∴ Ans $2.\overline{64}$

(4) Simplify : $(0.\bar{1})^2 \left(1 - 9 (0.1\bar{6})^2\right)$

a) $-1/162$

b) $1/108$

c) $7696/10^9$

D) 106

Sol^n Given :- $(0.\bar{1})^2\left(1-9(0.1\bar{6})^2\right)$

$$= \left(\frac{1}{9}\right)^2\left(1-9\left(\frac{16-1}{90}\right)^2\right)$$

$$= \frac{1}{81}\left(1-9\left(\frac{\cancel{15}^{\,1}}{\cancel{90}_{\,6}}\right)^2\right)$$

$$= \frac{1}{81}\left(1-9\left(\frac{1}{6}\right)^2\right)$$

$$= \frac{1}{81}\left(1-\cancel{9}^{\,1}\times\frac{1}{\cancel{36}_{\,4}}\right)$$

$$\frac{4-1}{4}$$

$$= \frac{1}{\cancel{81}_{\,27}} \times \frac{\cancel{3}^{\,1}}{4}$$

$$= \frac{1}{27\times 4} = \frac{1}{108}$$

$\therefore$ Ans b) $\frac{1}{108}$

5) Find the cube root of $0.0\overline{37}$:

a) $0.\bar{3}$

b) $0.\overline{13}$

c) $0.\overline{23}$

d) $0.2\bar{3}$

Sol^n :- $0.0\overline{37} = \frac{0037-0}{999}$ $\left(\because \frac{abcd-a}{999}\right)$

a b c d

$$= \frac{37}{999} = \frac{\cancel{37}\ 1}{9\times\cancel{111}\ 3}$$

Cube root $= \frac{1}{27}$

$$\sqrt[3]{\frac{1}{27}} = \frac{1}{3} = 0.333$$

$$= 0.\bar{3}$$

Ans a) $0.\bar{3}$

(5)

(2)

(5) If $0.a\bar{b} + 0.b\bar{a} = 7/9$ then how many pairs are possible for (a, b)?

a) 5
b) 4
c) 3
d) 1

$\because 0.a\bar{b} + 0.b\bar{a} = 7/9$

$$\frac{ab-a}{90} + \frac{ba-b}{90} = \frac{7}{9}$$

w.r.t

Ex:- put $ab = 23$ $a = 2$, $b = 3$

decimal, unit

10's, 1's. for 2 digit

for 3 digit xyz: thousand, decimal, unit.

$10a + b \times 1$

Ex. $10 \times 2 + 3 \times 1 = 20 + 3 = 23.$

$$\therefore \frac{ab-a}{90} + \frac{ba-b}{90} = \frac{7}{9}$$

$$\frac{10a + b - a}{90} + \frac{10b + a - b}{90} = \frac{7}{9}$$

$$\frac{10a + \cancel{b} - \cancel{a} + 10b + \cancel{a} - \cancel{b}}{90} = \frac{7}{9}$$

$$\frac{10a + 10b}{90} = \frac{7}{9}$$

$$\frac{\cancel{10}(a+b)}{\cancel{90}_{9}} = \frac{7}{9}$$

$$\frac{a+b}{\cancel{9}} = \frac{7}{\cancel{9}}$$

$$\boxed{a+b = 7}$$

$1 + 6 = 7$
$2 + 5 = 7$
$3 + 4 = 7$
$4 + 3 = 7$ → repeat.
$5 +$

$\therefore$ 3 times $\therefore$ Ans = c) 3.

(5)
(3)

(H.W)

Find the value of $0.\overline{57} - 0.4\overline{32} + 0.3\overline{5}$

a) $0.\overline{434}$

b) $0.\overline{497}$

c) $0.4\overline{98}$

d) $0.\overline{498}$

Solⁿ

$$
\begin{array}{r}
0.575757 \\
-\ 0.432323 \\
\hline
0.143434 \\
+\ 0.355555 \\
\hline
0.498989
\end{array}
$$

~~$0.\overline{57} - 0.4\overline{32} + 0.3\overline{5}$~~

$$
\begin{array}{r}
0.5777 \\
-\ 0.4322 \\
\hline
0.1455 \\
0.3555 \\
\hline
5.10
\end{array}
$$

$\therefore$ Ans :- 0.4989898

$0.4\overline{98}$